ST185 CELICA - CONNECTOR ATLAS

Every connector of all 15 harnesses, and every OEM junction box / plug / ground point we splice into.

Generated from docs/harness/wirelist/*.csv by docs/harness/atlas/build_atlas.py. Never edit the PDF; change the wire list and regenerate.

DRAWN	SHEETS	REV	SOURCE HASH	PROJECT
DG	13 (11 x 17 in landscape)	A	6ce22b6442a05bf0	st185-link-ecu-config

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HOW TO READ

Cavity circle: number inside, short signal label under it, fill = function (legend below). White = spare.
Connector title: name, maker part number, harness, true cavity count. "mates X" names the other half.
OEM boxes: outline of the factory box; each row = factory pin, factory wire colour, what the factory lead does, what our wire does and where it goes, EWD page and splice method.
Factory wire colours use the Toyota codes: B black, W white, R red, G green, L blue, Y yellow, O orange, BR brown, GR grey, P pink, LG light green; second letter = stripe.

Spare | CAN | Shield drain | +12 V power | Sensor signal | Output / actuator | Ground / sensor return | Sensor supply +5/+8 V | Coil trigger | Injector drive | Other / lighting / misc

Fill = function (keyword rule on the wire-list description). Circle size = contact size (DRB102 only). Only the DRB102 is its real maker insert face; every other connector is a numbered grid with the true cavity count (pin order, not the maker face view). Wire lists are the truth. Spare = no wire on that cavity.

ECU AND BULKHEAD (1 of 2)

DRB102 firewall bulkhead - insert face (TE drawing view)

DRB12-102PAE (cabin) / DRB16-102SAE (engine bay) | harnesses: A-cabin, A-engine, B-cabin, B-engine | circle size = contact size (4 > 8 > 12 > 16)

Left wedgelock half (52-102)

- 94 — CAN-Lambda shield
- 82
- 93 — CAN L to CAN-Lambda
- 81
- 70
- 102
- 92 — CAN H to CAN-Lambda
- 80
- 69
- 101
- 91 — Gearbox speed shield
- 79
- 57
- 68
- 62 — Back-up sw +12 V in
- 100
- 90 — +12 V gearbox speed
- 78 — Gnd fuel press + VSS
- 56 — EHPS relay coil (-)
- 67
- 61
- 99
- 89 — Gearbox speed signal
- 77 — Gnd oil/clnt/temps
- 55
- 60
- 98
- 88 — ETB motor shield
- 76 — Reverse sw +12 V out
- 54
- 66
- 59
- 97
- 87 — ETB motor -
- 75 — Charge temp signal
- 53
- 65
- 58
- 96 — CAN-Lambda ground
- 86 — ETB motor +
- 74 — Oil temp signal
- 52
- 64
- 95 — CAN-Lambda +12 V
- 85 — Knock sensor shield
- 73 — Coolant press signal
- 63
- 84 — Knock sensor (-)
- 72 — Fuel pressure signal
- 83 — Knock sensor signal
- 71 — Oil pressure signal

Right wedgelock half (1-51)

- 32 — Injector 2 drive (-)
- 20 — +12 V boost + flex
- 31 — Injector 1 drive (-)
- 19 — Turbo speed shield
- 40 — Coolant temp signal
- 30 — MAP sensor signal
- 18 — Gnd turbo speed
- 8
- 39 — Boost solenoid (-)
- 51 — +5 V coolant press
- 29 — Gnd flex + MAP
- 17 — Turbo speed signal
- 7
- 45
- 38 — Coil 4 trigger (IGT)
- 50 — +5 V fuel pressure
- 28 — +5 V MAP sensor
- 16 — +5 V turbo speed
- 6
- 44 — EHPS pump speed
- 37 — Coil 3 trigger (IGT)
- 49
- 27 — Throttle shield
- 15 — Cam sensor shield
- 5
- 43
- 48
- 26 — Gnd IAT/TPS/coolant
- 14 — Gnd cam sensor
- 4
- 42 — EHPS enable +12 V
- 36 — Coil 2 trigger (IGT)
- 47 — +5 V oil pressure
- 25 — Throttle pos 2 sub
- 13 — Cam sensor signal
- 3
- 41 — Intake temp signal
- 35 — Coil 1 trigger (IGT)
- 46
- 24 — Throttle pos 1 main
- 12 — +8 V to cam sensor
- 2 — Coil +12 V feed
- 34 — Injector 4 drive (-)
- 23 — +5 V throttle sensor
- 11 — Crank sensor shield
- 1 — Injector +12 V feed
- 33 — Injector 3 drive (-)
- 22 — Flex fuel shield
- 10 — Gnd crank sensor
- 21 — Flex fuel signal
- 9 — Crank sensor signal

ECU-A

4-1437290-1 | A-cabin | 34 cavities | numbered grid (true count)

A1 Injector 4 drive (-)	A2 Injector 3 drive (-)	A3 Injector 2 drive (-)	A4 Injector 1 drive (-)	A5 ECU main +12 V	A6 +8 V sensor supply	A7 Shield drain
A8 Crank sensor signal	A9 Cam sensor signal	A10 Coil 4 trigger (IGT)	A11 Coil 3 trigger (IGT)	A12 Coil 2 trigger (IGT)	A13 Coil 1 trigger (IGT)	A14 Pedal 1 signal main
A15 Coolant temp signal	A16 Intake temp signal	A17 MAP sensor signal	A18 A/C cancel (ACT)	A19 Fuel pump relay (-)	A20 ETB relay coil (-)	A21 Boost solenoid (-)
A22 Throttle pos 1 main	A23 FL conditioned out	A24 Sensor ground	A25 ECU power ground	A26 Start relay coil (-)	A27 EHPS pump speed	A28 EFI relay coil (-)
A29 Rad fan relay (-)	A30 Turbo speed signal	A31 Flex fuel signal	A32 +5 V sensor supply	A33 Throttle pos 2 sub	A34 ECU power ground	

Spare | CAN | Shield drain | +12 V power | Sensor signal | Output / actuator | Ground / sensor return | Sensor supply +5/+8 V | Coil trigger | Injector drive | Other / lighting / misc

Fill = function (keyword rule on the wire-list description). Circle size = contact size (DRB102 only). Only the DRB102 is its real maker insert face; every other connector is a numbered grid with the true cavity count (pin order, not the maker face view). Wire lists are the truth. Spare = no wire on that cavity.

ECU AND BULKHEAD (2 of 2)

ECU-B (Link G4X plug B, centre dash)

4-1437290-0 | B-cabin | 34 cavities | numbered grid (true count)

B1	B2	B3	B4	B5 ETB motor +12 V	B6 Oil temp signal	B7 Charge temp signal
B8	B9 Knock sensor signal	B10 A/C clutch relay (-)	B11	B12 EPS enable relay (-)	B13 Cond fan relay (-)	B14
B15 Oil pressure signal	B16 Fuel pressure signal	B17 Shield drain B	B18 ETB motor +	B19 Rear right wheel spd	B20 Rear left wheel spd	B21 FR conditioned out
B22 Sensor ground B	B23 Coolant press signal	B24 Fuel level signal	B25 ECU power ground	B26 ETB motor -	B27 Anti-theft input	B28 Ignition sw sense
B29 Gearbox speed signal	B30 Start request	B31 A/C request button	B32 Evap temp signal	B33 Pedal 2 signal sub	B34 ECU power ground	

ECU COMMS

DTM06-6S | CAN | 6 cavities | numbered grid (true count)

C1 CAN screen trunk	C2	C3 CAN 1 high	C4 CAN 1 low	C5	C6

Spare | CAN | Shield drain | +12 V power | Sensor signal | Output / actuator | Ground / sensor return | Sensor supply +5/+8 V | Coil trigger | Injector drive | Other / lighting / misc

Fill = function (keyword rule on the wire-list description). Circle size = contact size (DRB102 only). Only the DRB102 is its real maker insert face; every other connector is a numbered grid with the true cavity count (pin order, not the maker face view). Wire lists are the truth. Spare = no wire on that cavity.

CABIN

CSB3
HD36-24-33SE | CSB3 | 33 cavities | numbered grid (true count)

1	2	3	4	5	6	7
CSB3 +12 V	CSB3 ground	Opto +5 V	Opto out ground	CAN H to CSB3	CAN L to CSB3	
8	9	10	11	12	13	14
Cruise ladder						
15	16	17	18	19	20	21
		Cruise MAIN on			Clutch pedal sw	Brake to CSB3 S7
22	23	24	25	26	27	28
Reverse to CSB3 S8				Oil LED sink (CSB3)		
29	30	31	32	33		

ALARM
DT06-12SA | AntiTheft | 12 cavities | numbered grid (true count)

1	2	3	4	5	6
Alarm +12 V	Alarm ground	Alarm ignition +12V	Siren +	Park lamp flash	Door trigger
7	8	9	10	11	12
Hood trigger	Trunk trigger	Anti-theft input	Status LED +	Valet switch	Shock trigger

OPTO8
(Dan's on-hand board - generic 8-ch EL814 opt... | CSB3 | 20 cavities | numbered grid (true count)

IN2	VCC	GND	OUT1	OUT2	IN1	COM1
Reverse to opto in 2	Opto +5 V	Opto out ground	Brake to CSB3 S7	Reverse to CSB3 S8	Brake on +12 V	Opto COM ground
COM2	IN3	IN4	IN5	IN6	IN7	IN8
OUT3	OUT4	OUT5	OUT6	OUT7	OUT8	

FRONT WS INLINE
DT06-6S | A-cabin | 6 cavities | numbered grid (true count)

1	2	3	4	5	6
VRC-F +5V	VRC-F GND	FL conditioned out	FR conditioned out	WS-F screen drain	

mates ix_ws_front_r on WheelSpeed-Front

PEDAL
Sumitomo TS 025 6-way | A-cabin | 6 cavities | numbered grid (true count)

1	2	3	4	5	6
+5 V pedal sensor 2	Pedal 2 ground	Pedal 2 signal sub	+5 V pedal sensor 1	Pedal 1 ground	Pedal 1 signal main

+5V BUS
DT06-08SA | A-cabin | 8 cavities | numbered grid (true count)

1	2	3	4	5	6	7	8
+5 V sensor supply	+5 V throttle sensor	+5 V MAP sensor	+5 V turbo speed	+5 V oil pressure	+5 V fuel pressure	+5 V coolant press	

mates bus_5v_bha_r

+5V BUS cap
DT04-08PA-P021 | A-cabin | 8 cavities | numbered grid (true count)

1	2	3	4	5	6	7	8
+5 V sensor supply	+5 V throttle sensor	+5 V MAP sensor	+5 V turbo speed	+5 V oil pressure	+5 V fuel pressure	+5 V coolant press	

mates bus_5v_bha_p

SENSOR GND BUS
DT06-08SA | A-cabin | 8 cavities | numbered grid (true count)

1	2	3	4	5	6	7	8
Sensor ground	Gnd IAT/TPS/coolant	Gnd crank sensor	Gnd cam sensor	Gnd turbo speed	Gnd flex + MAP		

mates bus_gnd_bha_r

SENSOR GND BUS cap
DT04-08PA-P021 | A-cabin | 8 cavities | numbered grid (true count)

1	2	3	4	5	6	7	8
Sensor ground	Gnd IAT/TPS/coolant	Gnd crank sensor	Gnd cam sensor	Gnd turbo speed	Gnd flex + MAP		

mates bus_gnd_bha_p

SHOCK SNSR
DT06-3S | AntiTheft | 3 cavities | numbered grid (true count)

B	C	A
Shock sensor ground	Shock trigger	Shock sensor +12 V

SIREN
DT06-2S-CE05 | AntiTheft | 2 cavities | numbered grid (true count)

1	2
Siren +	Siren ground

STATUS LED
DT06-2S | AntiTheft | 2 cavities | numbered grid (true count)

1	2
Status LED +	Status LED gnd

VALET SW
DT06-2S | AntiTheft | 2 cavities | numbered grid (true count)

1	2
Valet switch	Valet switch ground

IX-FUEL inline to fuel sender lead...
DT04-2P | B-cabin | 2 cavities | numbered grid (true count)

1	2
Fuel level signal	Fuel level ground

mates ix_fuel_p on RearFuel

IX-WS-R inline to rear VRC (dash above ECU)
DT06-6S | B-cabin | 6 cavities | numbered grid (true count)

1	2	3	4	5	6
VRC-R +5V	VRC-R GND	Rear left wheel spd	Rear right wheel spd	WS-R screen drain	

mates ix_ws_rear_r on WheelSpeed-Rear

CAN-LAMBDA
DTM06-4S | CAN | 4 cavities | numbered grid (true count)

1	2	3	4
CAN-Lambda +12 V	CAN-Lambda ground	CAN L to CAN-Lambda	CAN H to CAN-Lambda

Spare | CAN | Shield drain | +12 V power | Sensor signal | Output / actuator | Ground / sensor return | Sensor supply +5/+8 V | Coil trigger | Injector drive | Other / lighting / misc

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ENGINE BAY (1 of 2)

EHPS PUMP
MR-S ZZW30 EHPS pump - A 90980-12068 / B 9098... | EngineRoom-C | 10 cavities | numbered grid (true count)

A-1	C-1	B-2	B-5	A-2
EHPS pump +12 V	EHPS enable +12 V	EHPS pump speed	EHPS relay coil (-)	EHPS pump ground

B-1	B-3	B-4	B-6	C-2

Turbo speed sensor
DT06-3S-CE05 | A-engine | 3 cavities | numbered grid (true count)

A	B	C
+5 V turbo speed	Turbo speed signal	Gnd turbo speed

MAP sensor (sensor block, LH firewall)
12110293 | A-engine | 3 cavities | numbered grid (true count)

1	2	3
Gnd MAP sensor	+5 V MAP sensor	MAP sensor signal

Flex fuel sensor (fuel return line)
13519047 | A-engine | 3 cavities | numbered grid (true count)

C	A	B
Flex fuel signal	+12 V flex fuel	Gnd flex fuel

Injector cyl 1
15419715 | A-engine | 2 cavities | numbered grid (true count)

1	2
Injector 1 +12 V	Injector 1 drive (-)

Injector cyl 2
15419715 | A-engine | 2 cavities | numbered grid (true count)

1	2
Injector 2 +12 V	Injector 2 drive (-)

Injector cyl 3
15419715 | A-engine | 2 cavities | numbered grid (true count)

1	2
Injector 3 +12 V	Injector 3 drive (-)

Injector cyl 4
15419715 | A-engine | 2 cavities | numbered grid (true count)

1	2
Injector 4 +12 V	Injector 4 drive (-)

Ignition coil cyl 1
90980-11885 | A-engine | 4 cavities | numbered grid (true count)

1	2	3	4
Coil 1 +12 V		Coil 1 trigger (IGT)	Coil 1 ground

Ignition coil cyl 2
90980-11885 | A-engine | 4 cavities | numbered grid (true count)

1	2	3	4
Coil 2 +12 V		Coil 2 trigger (IGT)	Coil 2 ground

Ignition coil cyl 3
90980-11885 | A-engine | 4 cavities | numbered grid (true count)

1	2	3	4
Coil 3 +12 V		Coil 3 trigger (IGT)	Coil 3 ground

Ignition coil cyl 4
90980-11885 | A-engine | 4 cavities | numbered grid (true count)

1	2	3	4
Coil 4 +12 V		Coil 4 trigger (IGT)	Coil 4 ground

Throttle body (Bosch ETB)
D 261 205 358-01 | A-engine | 6 cavities | numbered grid (true count)

1	2	3	4	5	6
ETB motor -	Gnd throttle sensor	+5 V throttle sensor	ETB motor +	Throttle pos 2 sub	Throttle pos 1 main

Boost control solenoid
12052641 | A-engine | 2 cavities | numbered grid (true count)

1	2
+12 V boost sol	Boost solenoid (-)

Coolant temp sensor (water neck)
90980-11062 | A-engine | 2 cavities | numbered grid (true count)

1	2
Coolant temp signal	Gnd coolant temp

Intake air temp sensor (manifold)
90980-11062 | A-engine | 2 cavities | numbered grid (true count)

1	2
Intake temp signal	Gnd intake temp

Crank sensor
DT06-2S-CE05 | A-engine | 2 cavities | numbered grid (true count)

1	2
Crank sensor signal	Gnd crank sensor

Cam sensor (Hall)
(kit-supplied pigtail) | A-engine | 3 cavities | numbered grid (true count)

1	2	3
+8 V to cam sensor	Cam sensor signal	Gnd cam sensor

IX-B-RAIL inline plug, +5 V to B-engine
DT06-6S-CE05 | A-engine | 6 cavities | numbered grid (true count)

1	2	3	4	5	6
+5 V oil pressure	+5 V fuel pressure	+5 V coolant press			

mates ix_brail_p on B-engine

Oil pressure sensor (sensor block, LH...
12110293 | B-engine | 3 cavities | numbered grid (true count)

1	2	3
Gnd oil pressure	+5 V oil pressure	Oil pressure signal

Fuel pressure sensor (sensor block, L...
12110293 | B-engine | 3 cavities | numbered grid (true count)

1	2	3
Gnd fuel pressure	+5 V fuel pressure	Fuel pressure signal

Coolant pressure sensor (rear of block...
12110293 | B-engine | 3 cavities | numbered grid (true count)

1	2	3
Gnd coolant press	+5 V coolant press	Coolant press signal

Oil temp sender
90980-11062 | B-engine | 2 cavities | numbered grid (true count)

1	2
Oil temp signal	Gnd oil temp

Back-up light switch B1 (transaxle)
90980-11250 | B-engine | 2 cavities | numbered grid (true count)

1	2
Reverse sw +12 V out	Back-up sw +12 V in

Charge-pipe air temp sensor
90980-11062 | B-engine | 2 cavities | numbered grid (true count)

1	2
Charge temp signal	Gnd charge temp

Gearbox speed sensor (transaxle)
90980-11143 | B-engine | 3 cavities | numbered grid (true count)

1	2	3
+12 V gearbox speed	Gnd gearbox speed	Gearbox speed signal

Knock sensor 1 (block)
1 928 403 874 | B-engine | 2 cavities | numbered grid (true count)

1	2
Knock sensor signal	Knock sensor (-)

IX-B-RAIL inline socket, +5 V from A-engine
DT04-6P-CE03 | B-engine | 6 cavities | numbered grid (true count)

1	2	3	4	5	6
+5 V oil pressure	+5 V fuel pressure	+5 V coolant press			

mates ix_brail_s on A-engine

POWER BUS
ST185-FB-POWER rev 7 BUS ZONE | EngineRoom-C | 3 cavities | numbered grid (true count)

GND	BAT+	ALT/STR
POWER box gnd to ID / POWER box gnd to IG / Engine...	Battery + to POWER	Starter feed (H3)

RAD FAN+
DTHD06-1-8S | EngineRoom-C | 1 cavities | numbered grid (true count)

+
Rad fan +12 V

COND FAN+
DTHD06-1-8S | EngineRoom-C | 1 cavities | numbered grid (true count)

+
Cond fan +12 V

RAD FAN-
DTHD06-1-8S | EngineRoom-C | 1 cavities | numbered grid (true count)

-
Rad fan ground

COND FAN-
DTHD06-1-8S | EngineRoom-C | 1 cavities | numbered grid (true count)

-
Cond fan ground

HORN LH
DT06-2S-CE05 | EngineRoom-C | 2 cavities | numbered grid (true count)

1	2
Horn LH +12 V	Horn LH ground

HORN RH
DT06-2S-CE05 | EngineRoom-C | 2 cavities | numbered grid (true count)

1	2
Horn RH +12 V	Horn RH ground

Spare · CAN · Shield drain · +12 V power · Sensor signal · Output / actuator · Ground / sensor return · Sensor supply +5/+8 V · Coil trigger · Injector drive · Other / lighting / misc

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ENGINE BAY (2 of 2)

WSS FL

DT06-2S-CE05 | WheelSpeed-Front | 2 cavities | numbered grid (true count)

1	FL wheel speed +
2	FL wheel speed -

WSS FR

DT06-2S-CE05 | WheelSpeed-Front | 2 cavities | numbered grid (true count)

1	FR wheel speed +
2	FR wheel speed -

IX-WS-F

DT04-6P | WheelSpeed-Front | 6 cavities | numbered grid (true count)

1	VRC-F +5V
2	VRC-F GND
3	FL conditioned out
4	FR conditioned out
5	WS-F screen drain
6	

mates ix_ws_front_p on A-cabin

VRC-F IN

LP-16-J05PE-02-001 | WheelSpeed-Front | 5 cavities | numbered grid (true count)

1	FL wheel speed +
2	FL wheel speed -
3	Front sensor shields
4	FR wheel speed +
5	FR wheel speed -

VRC-F OUT

LP-16-J05PE-02-001 | WheelSpeed-Front | 5 cavities | numbered grid (true count)

1	VRC-F +5V
2	VRC-F GND
3	FL conditioned out
4	FR conditioned out
5	WS-F screen drain

Spare | CAN | Shield drain | +12 V power | Sensor signal | Output / actuator | Ground / sensor return | Sensor supply +5/+8 V | Coil trigger | Injector drive | Other / lighting / misc

Fill = function (keyword rule on the wire-list description). Circle size = contact size (DRB102 only). Only the DRB102 is its real maker insert face; every other connector is a numbered grid with the true cavity count (pin order, not the maker face view). Wire lists are the truth. Spare = no wire on that cavity.

FUSE BOXES, RELAYS AND HEAVY DC (1 of 2)

FB-LOGIC

ST185-FB-LOGIC rev 5 | CabinPower | 25 cavities | numbered grid (true count)

BUS2-12	BUS2-13	BUS2-5	BUS1-1	BUS1-10	BUS1-2	BUS1-3
EHPS enable +12 V	Start request	CSB3 +12 V	EFI relay coil+	ETB relay +12 V in	Head LH +12 V feed	Head RH +12 V feed
BUS1-4	BUS1-7	BUS1-8	BUS2-1	BUS2-2	BUS2-3	BUS2-4
Haz-horn +12 V feed	Alarm +12 V	Shock sensor +12 V	CAN-Lambda +12 V	Switched +12 V feed	Coil +12 V feed	Injector +12 V feed
BUS2-6	BUS2-7	BUS2-8	BUS2-9	BUS1	BUS2	BUS2-12 IN
Dash devices +12 V	EPS enable +12 V in	A/C clutch +12 V in	Alarm ignition +12V	LOGIC Bus 1 feed	LOGIC Bus 2 feed	EPS enable out
BUS2-13 IN	BUS1-5	BUS1-6	BUS1-9			
Start sw ST1 feed	DOME feed to J/B1	Door lock feed	Alt S battery sense			

FB-POWER

ST185-FB-POWER rev 7 | CabinPower | 14 cavities | numbered grid (true count)

AMI IN-1	AMI IN-2	AMI IN-3	AMI IN-4	AMI IN-5	AMI IN-6	AMI OUT-1
Heater relay feed	EFI relay feed	Start relay feed	Rad fan relay feed	Cond fan relay feed	Power relay feed	LOGIC Bus 1 feed
AMI OUT-6	ACPS IN	ACPS	AMI OUT-2	AMI OUT-3	AMI OUT-4	AMI OUT-5
EHPS relay feed	ACPS fuse feed	A/C press sw feed	Retract relay feed	AM1 feed to IE1	AM2 feed to IE1	J/B1 main feed

SW12 bus plug DT06-12 (glove box, by LOGIC box)

DT06-12SA | CabinPower | 12 cavities | numbered grid (true count)

1	2	3	4	5	6
Switched +12 V feed	Switched +12 V feed	Fuel pump rly coil+	Rad fan rly coil+	ETB relay coil+	Start relay coil+
7	8	9	10	11	12
Cond fan rly coil+	EHPS relay coil+	EPS enable coil+			

mates bus_sw12_r

SW12 bussed cap DT04-12 P021 (on plug)

DT04-12PA-P021 | CabinPower | 12 cavities | numbered grid (true count)

1	2	3	4	5	6
Switched +12 V feed	Switched +12 V feed	Fuel pump rly coil+	Rad fan rly coil+	ETB relay coil+	Start relay coil+
7	8	9	10	11	12
Cond fan rly coil+	EHPS relay coil+	EPS enable coil+			

mates bus_sw12_p

POWER R2

1-1414147-0 | CabinPower | 4 cavities | numbered grid (true count)

30	85	86	87
LOGIC Bus 2 feed	EFI relay coil (-)	EFI relay coil+	EFI relay feed

POWER R4

1-1414147-0 | CabinPower | 4 cavities | numbered grid (true count)

30	85	86	87
Rad fan +12 V	Rad fan relay (-)	Rad fan rly coil+	Rad fan relay feed

LOGIC M4

4-1904124-2 | CabinPower | 5 cavities | numbered grid (true count)

86	30	85	87	87a
ETB relay coil+	ETB relay +12 V in	ETB relay coil (-)	ETB motor +12 V	

POWER R3

1-1414147-0 | CabinPower | 4 cavities | numbered grid (true count)

30	85	86	87
Starter solenoid S	Start relay coil (-)	Start relay coil+	Start relay feed

POWER R5

1-1393304-0 | CabinPower | 4 cavities | numbered grid (true count)

30	85	86	87
Cond fan +12 V	Cond fan relay (-)	Cond fan rly coil+	Cond fan relay feed

EHPS RELAY

V23132-A2001-B200 | CabinPower | 5 cavities | numbered grid (true count)

85	86	30	87	87a
EHPS relay coil (-)	EHPS relay coil+	EHPS relay feed	EHPS pump +12 V	

LOGIC M5

6-1419137-4 | CabinPower | 5 cavities | numbered grid (true count)

86	85	30	87a	87
A/C clutch relay (+)	A/C clutch relay (-)	A/C clutch +12 V in		A/C clutch +12 V

LOGIC M6

6-1419137-4 | CabinPower | 5 cavities | numbered grid (true count)

85	86	30	87	87a
EPS enable relay (-)	EPS enable coil+	EPS enable +12 V in	EPS enable out	

LOGIC M7

6-1419137-4 | CabinPower | 5 cavities | numbered grid (true count)

86	85	87	30	87a
Reverse relay coil+	Reverse relay coil-	Back-up lamps +12 V	Reverse relay feed	

LOGIC M1

5-1393292-8 | CabinPower | 5 cavities | numbered grid (true count)

85	30	86	87a	87
Head LH coil (-)	Head LH relay feed	Head LH relay coil+		Head LH +12 V out

LOGIC M2

5-1393292-8 | CabinPower | 5 cavities | numbered grid (true count)

85	30	86	87a	87
Head RH coil (-)	Head RH relay feed	Head RH relay coil+		Head RH +12 V out

LOGIC M3

4-1904124-3 | CabinPower | 5 cavities | numbered grid (true count)

85	87	30	86	87a
Horn relay coil (-)	Horn +12 V out	Horn relay feed	Horn relay coil+	

POWER R1

1-1414687-0 | CabinPower | 5 cavities | numbered grid (true count)

87	30	85	87a	86
Heater relay feed	Heater relay out	Heater relay coil-	Heater relay NC gnd	Heater relay coil+

POWER R6

1-1393304-0 | CabinPower | 4 cavities | numbered grid (true count)

30	85	86	87
Power window feed	Power relay coil-	Power relay coil+	Power relay feed

RADLOK+ FW

RL00801-50RE | HeavyDC | 1 cavities | numbered grid (true count)

1
Starter feed (H3)

mates rl_pos_eng on HeavyDC

Spare | CAN | Shield drain | +12 V power | Sensor signal | Output / actuator | Ground / sensor return | Sensor supply +5/+8 V | Coil trigger | Injector drive | Other / lighting / misc

Fill = function (keyword rule on the wire-list description). Circle size = contact size (DRB102 only). Only the DRB102 is its real maker insert face; every other connector is a numbered grid with the true cavity count (pin order, not the maker face view). Wire lists are the truth. Spare = no wire on that cavity.

FUSE BOXES, RELAYS AND HEAVY DC (2 of 2)

RADLOK+ ENG

RL00801-50RE | HeavyDC | 1 cavities | numbered grid (true count)

1
Starter feed (H3)

mates rl_pos_fw on HeavyDC

RADLOK- FW

RL00801-50BK | HeavyDC | 1 cavities | numbered grid (true count)

1
Engine ground (H6a)

mates rl_neg_eng on HeavyDC

RADLOK- ENG

RL00801-50BK | HeavyDC | 1 cavities | numbered grid (true count)

1
Engine ground (H6a)

mates rl_neg_fw on HeavyDC

JUMP POST

jump-post-M8 | HeavyDC | 1 cavities | numbered grid (true count)

1
Jump post feed

ALT PLUG G2

90980-10341 | HeavyDC | 3 cavities | numbered grid (true count)

1	2	3
Alt S battery sense	Alt L charge lamp	Alt IG excite

KILL SWITCH

52-45780 | HeavyDC | 4 cavities | numbered grid (true count)

ALT 2	MAIN 2	ALT 1	MAIN 1
Alt IG excite	Battery relay box + / Battery + to POWER	Alt IG to kill sw	Battery + main

BATT RLY BOX

ST185-FB-BATT rev 5 | HeavyDC | 2 cavities | numbered grid (true count)

FP 40A	BATT +
Fuel pump rly feed	Battery relay box +

FP RELAY K1

V23132-A2001-B200 | HeavyDC | 5 cavities | numbered grid (true count)

85	86	30	87a	87
Fuel pump relay (-)	Fuel pump rly coil+	Fuel pump rly feed		Fuel pump +12 V

Spare · CAN · Shield drain · +12 V power · Sensor signal · Output / actuator · Ground / sensor return · Sensor supply +5/+8 V · Coil trigger · Injector drive · Other / lighting / misc

Fill = function (keyword rule on the wire-list description). Circle size = contact size (DRB102 only). Only the DRB102 is its real maker insert face; every other connector is a numbered grid with the true cavity count (pin order, not the maker face view). Wire lists are the truth. Spare = no wire on that cavity.

OEM JUNCTION BOXES AND SPLICE POINTS (1 of 4)

Body ground points we use (1 of 2)

factory ground studs and bolts; each gets a ring terminal

Cabin ground stud near the alarm

GND B	spot and EWD ID TBD, M6	**our w_at_gnd_a1: 20 AWG Black** from ALARM pin 2 ring terminal on the factory ground / stud
GND B	spot and EWD ID TBD, M6	**our w_at_gnd_b2: 20 AWG Black** from STATUS LED pin 2 ring terminal on the factory ground / stud
GND B	spot and EWD ID TBD, M6	**our w_at_gnd_b1: 20 AWG Black** from SHOCK SNSR pin B ring terminal on the factory ground / stud
GND B	spot and EWD ID TBD, M6	**our w_at_gnd_a2: 20 AWG Black** from SIREN pin 2 ring terminal on the factory ground / stud
GND B	spot and EWD ID TBD, M6	**our w_at_gnd_c1: 20 AWG Black** from VALET SW pin 2 ring terminal on the factory ground / stud

Engine block ground stud

GND B	RADLOK engine ground, H6	**our w_eng_gnd: 10 AWG Black** from splice sp_chassis_eng (Coil ground splice) ring terminal on the factory ground / stud

Engine block ground stud, lug 1

GND B	Route 1 point 5	**our w_rln_blk: 53.5 AWG Black** from RADLOK- ENG pin 1 ring terminal on the factory ground / stud

Engine block ground stud, lug 2

GND B	Route 1 point 5	**our w_gnd_eb: 4 AWG Black** from Ground stud EB, LH front fender (EWD ground point... ring terminal on the factory ground / stud

Engine block ground stud, lug 3

GND B	Route 1 point 5	**our w_eps_gnd: 8 AWG Black** from EHPS PUMP pin A-2 ring terminal on the factory ground / stud

Ground bolt IG, right kick

GND B	EWD ground point IG	**our w_gnd_ig: 53.5 AWG Black** from POWER BUS pin GND ring terminal on the factory ground / stud

Ground point BL, back panel center

GND B	EWD 1993 p.112	**our w_rl_gnd_bl: 16 AWG White** from splice sp_rl_gnd (Rear ground splice) ring terminal on the factory ground / stud

Ground point IF, I/P brace RH

GND B	EWD ground point IF, p.42	**our w_gnd_clu: 16 AWG Black** from Cluster buck converter IN- pin (old meter spot) ring terminal on the factory ground / stud
GND B	EWD ground point IF, p.42	**our w_gnd_rd: 16 AWG Black** from Pi 12-to-5 V supply IN- pin (center dash) ring terminal on the factory ground / stud

Ground stud EA, RH front fender

GND B	EWD ground point EA	**our w_fan2_gnd: 8 AWG Black** from COND FAN- pin - ring terminal on the factory ground / stud
GND B	EWD ground point EA	**our w_horn_lh_g: 16 AWG Black** from HORN LH pin 2 ring terminal on the factory ground / stud
GND B	EWD ground point EA	**our w_horn_rh_g: 16 AWG Black** from HORN RH pin 2 ring terminal on the factory ground / stud

IE1 cowl plug

left kick panel, cowl side; engine-room main REMOVED, we tap the cowl-side leads

Leads / pins

2 R-Y	headlight relay coil line	**our w_head_coil_cp: 18 AWG Violet** to splice sp_head_coil (Headlight coil splice) (Headlight coil (-)) EWD 1993 p.116; crimp butt splice into the factory cowl-side lead (ca...
3 W	RTR 30 A to retract relay	**our w_pmu_rtr: 10 AWG Red** to FB-POWER pin AMI OUT-2 (Retract relay feed) EWD 1993 p.116; heavy crimp butt splice into the factory cowl-side le...
6 L-W	park switch S (EWD p.148) / OEM IE1 pin 6 (left kick, cowl side) - L-W, wiper...	**our w_wip_park: 14 AWG Blue** to Front wiper motor F9 pin 3 (cowl, driver side) (Wiper park sw) EWD 1993 p.148 / 1993 p.148; new-wire run terminated at the device's own mating pl...
7 G-Y	RH retract motor pin 4	**our w_rtr_r3_4: 12 AWG Green** to splice sp_rtr_r3_4 (RH retract motor feed) (RH retract motor 4) EWD 1993 p.117; crimp butt splice into the factory cowl-side lead (ca...
8 G-B	RH retract motor pin 2 (EWD p.117) / RH retract motor R3 pin 2 - motor feed f...	**our w_rtr_r3_2: 16 AWG Green** to RH retract motor R3 pin 2 (RH retract motor 2) EWD 1993 p.117 / 1993 p.117; crimp butt splice into the factory cowl-side lead (ca...
9 G-Y	retract close limit	**our w_rtr_close: 18 AWG Green** to splice sp_rtr_close (Retract close-limit join) (Retract close limit) EWD 1993 p.117; crimp butt splice into the factory cowl-side lead (ca...
10 W	AM1 to ignition switch	**our w_pdb_am1: 8 AWG White** to FB-POWER pin AMI OUT-3 (AM1 feed to IE1) EWD p.31; heavy crimp butt splice into the factory lead
11 R-L	high-beam return	**our w_hl_high: 14 AWG Red** to splice sp_hl_high (High-beam return join) (High-beam return) EWD 1993 p.116; crimp butt splice into the factory cowl-side lead (ca...
12 W	hazard / flasher feed	**our w_haz_cp: 14 AWG Red** to splice sp_hz (HAZ-HORN splice) (Hazard feed) EWD 1993 p.130; crimp butt splice into the factory cowl-side lead (ca...
13 L-B	wiper LOW (EWD p.148) / Front wiper motor F9 pin 2 (cowl, driver side) - LOW	**our w_wip_lo: 14 AWG Blue** to Front wiper motor F9 pin 2 (cowl, driver side) (Wiper LOW) EWD 1993 p.148 / 1993 p.148; crimp butt splice into the factory cowl-side lead (ca...
14 G	LH retract motor pin 4	**our w_rtr_r2_4: 12 AWG Green** to splice sp_rtr_r2_4 (LH retract motor feed) (LH retract motor 4) EWD 1993 p.117; crimp butt splice into the factory cowl-side lead (ca...
15 G-O	LH retract motor pin 2 (EWD p.117) / LH retract motor R2 pin 2 - motor feed f...	**our w_rtr_r2_2: 16 AWG Green** to LH retract motor R2 pin 2 (LH retract motor 2) EWD 1993 p.117 / 1993 p.117; crimp butt splice into the factory cowl-side lead (ca...
16 G-W	retract open limit	**our w_rtr_open: 18 AWG Green** to splice sp_rtr_open (Retract open-limit join) (Retract open limit) EWD 1993 p.117; crimp butt splice into the factory cowl-side lead (ca...
17 B-R	AM2 to ignition switch	**our w_pdb_am2: 10 AWG Brown** to FB-POWER pin AMI OUT-4 (AM2 feed to IE1) EWD p.31; heavy crimp butt splice into the factory lead
18 R-W	low-beam return	**our w_hl_low: 14 AWG Red** to splice sp_hl_low (Low-beam return join) (Low-beam return) EWD 1993 p.116; crimp butt splice into the factory cowl-side lead (ca...
19 G	horn switch line	**our w_horn_85_cp: 18 AWG Violet** to LOGIC M3 pin 85 (Horn relay coil (-)) EWD 1993 p.185; crimp butt splice into the factory cowl-side lead (ca...
20 L-O	wiper HIGH (EWD p.148) / Front wiper motor F9 pin 1 (cowl, driver side) - HIGH	**our w_wip_hi: 14 AWG Blue** to Front wiper motor F9 pin 1 (cowl, driver side) (Wiper HIGH) EWD 1993 p.148 / 1993 p.148; crimp butt splice into the factory cowl-side lead (ca...

J/B No.1 junction block

left kick panel; factory box KEPT; we splice factory leads of plugs 1F / 1H / 1I; cabin side

Plug 1F

1 G	TAIL 15A out	**our w_rl_tail: 16 AWG Green** to splice sp_rl_tail (Tail splice) (Tail lamp feed) EWD 1993 p.112-113; crimp butt splice into the factory 1F-1 lead at J/B1...
2 G-Y	turn RH out (EWD 1993 p.130-131) / RH rear lamp R12 pin 6, socket pigtail - t...	**our w_rl_turn_r: 18 AWG Green** to RH rear lamp R12 pin 6, socket pigtail (Turn RH rear +) EWD 1993 p.130-131 / 1993 p.130; crimp butt splice into the factory 1F-2 lead at J/B1...
3 G-B	turn LH out (EWD 1993 p.130-131) / LH rear lamp R11 pin 6, socket pigtail - t...	**our w_rl_turn_l: 18 AWG Green** to LH rear lamp R11 pin 6, socket pigtail (Turn LH rear +) EWD 1993 p.130-131 / 1993 p.130; crimp butt splice into the factory 1F-3 lead at J/B1...

Plug 1H

1 B	J/B1 always-hot B+	**our w_pdb_jb1: 6 AWG White** to FB-POWER pin AMI OUT-5 (J/B1 main feed) EWD p.18-19 / p.46; heavy crimp butt splice into the factory 1I-1 lead
2 L-Y	DOME 20 A feed	**our w_pmu_dome: 14 AWG Red** to FB-LOGIC pin BUS1-5 (DOME feed to J/B1) EWD 1993 p.132; crimp butt splice into the factory 1H-2 lead at J/B1...
4 G-B	turn LH out (EWD p.130) / LH front turn / clearance lamp F7 pin 3 - turn +	**our w_turn_f7: 18 AWG Green** to LH front turn / clearance lamp F7 pin 3 (LH front turn +) EWD 1993 p.130 / 1993 p.130; crimp butt splice into the factory 1H-4 lead at J/B1...
5 L	WIPER 20 A out (EWD p.148) / Front wiper motor F9 pin 4 (cowl, driver side) - +B	**our w_wip_b: 14 AWG Blue** to Front wiper motor F9 pin 4 (cowl, driver side) (Wiper +B) EWD 1993 p.148 / 1993 p.148; crimp butt splice into the factory 1H-5 lead at J/B1...
6 G	TAIL 15 A out	**our w_clr: 16 AWG Green** to splice sp_clr (Clearance feed split) (Clearance lamp feed) EWD 1993 p.112; crimp butt splice into the factory 1H-6 lead at J/B1...
7 B-O	IGN fuse out (EWD 1993 p.62); same lead as the ECU-B B28 sense tap, own splice	**our w_ig2_in: 20 AWG Green** to IG2 wake diode (1N5822, anode) (IG2 wake feed) EWD 1993 TIS EWD p.62 (Power Source); solder splice onto the factory lead
7 B-O	IGN fuse out	**our w_ign: 20 AWG Green** to ECU-B (Link G4X plug B, centre dash) pin B28 (Ignition sw sense) EWD 1993 TIS EWD p.62 (Power Source); solder splice onto the factory lead
8 G-Y	turn RH out (EWD p.130) / RH front turn / clearance lamp F8 pin 3 - turn +	**our w_turn_f8: 18 AWG Green** to RH front turn / clearance lamp F8 pin 3 (RH front turn +) EWD 1993 p.130 / 1993 p.130; crimp butt splice into the factory 1H-8 lead at J/B1...

Spare · CAN · Shield drain · +12 V power · Sensor signal · Output / actuator · Ground / sensor return · Sensor supply +5/+8 V · Coil trigger · Injector drive · Other / lighting / misc

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OEM JUNCTION BOXES AND SPLICE POINTS (2 of 4)

Body ground points we use (2 of 2)

factory ground studs and bolts; each gets a ring terminal

Ground stud EA, RH front fender

GND [redacted] B	EWD ground point EA	**our w_fan_gnd: 8 AWG Black** from RAD FAN- pin - ring terminal on the factory ground / stud
GND [redacted] B	EWD ground point EA	**our w_f8_gnd: 16 AWG White** from RH front turn / clearance lamp F8 pin 2 ring terminal on the factory ground / stud

Ground stud EB, LH front fender

GND [redacted] B	EWD ground point EB	**our w_f7_gnd: 16 AWG White** from LH front turn / clearance lamp F7 pin 2 ring terminal on the factory ground / stud
GND [redacted] B	EWD ground point EB	**our w_rtr_gnd_r2_t: 12 AWG White** from splice sp_rtr_gnd_r2 (LH retract motor ground) ring terminal on the factory ground / stud
GND [redacted] B	EWD ground point EB	**our w_rtr_gnd_r3_t: 12 AWG White** from splice sp_rtr_gnd_r3 (RH retract motor ground) ring terminal on the factory ground / stud
GND [redacted] B	EWD ground point EB	**our w_gnd_eb: 4 AWG Black** from Engine block ground stud, lug 2 (Route 1 point 5) ring terminal on the factory ground / stud

Ground stud ID, left kick panel

GND [redacted] B	EWD ground point ID	**our w_gnd_id: 53.5 AWG Black** from POWER BUS pin GND ring terminal on the factory ground / stud

Ground stud, LH kick panel

GND [redacted] B	EWD ground point ID	**our w_led_gnd_out: 20 AWG Black** from splice sp_led_gnd (LED ground splice) ring terminal on the factory ground / stud

Ground stud, trunk LH side

GND [redacted] B	EWD ID TBD, M6 assumed	**our w_fpg: 12 AWG Black** from FUEL PUMP pin 2 ring terminal on the factory ground / stud

LH rear lamp R11

R11 | device plug / lead tapped by our harness

1 G-W	stop LH	**our w_rl_stop_r11_c1: 16 AWG Green** to splice sp_rl_stop (Stop splice) (Stop LH +) EWD 1993 p.126; new-wire flying lead, soldered by Dan to the factory...
2 G	tail LH	**our w_rl_tail_r11_c2: 16 AWG Green** to splice sp_rl_tail (Tail splice) (Tail LH +) EWD 1993 p.113; new-wire flying lead, soldered by Dan to the factory...
3 G	rear side marker LH	**our w_rl_tail_r11_c3: 16 AWG Green** to splice sp_rl_tail (Tail splice) (Rear marker LH +) EWD 1993 p.113; new-wire flying lead, soldered by Dan to the factory...
4 R-B	back-up LH (Mitchell 1993 Fig. 16)	**our w_rl_rev_r11_c4: 16 AWG Red** to splice sp_rl_rev (Back-up splice) (Back-up LH +) EWD Mitchell 1993 Fig. 16 grid 63; 1992 p.132; new-wire flying lead, soldered by Dan to the factory...
5 W-B	lamp ground LH	**our w_rl_gnd_r11_c5: 16 AWG White** to splice sp_rl_gnd (Rear ground splice) (Rear lamp LH gnd) EWD 1993 p.113 / p.126; new-wire flying lead, soldered by Dan to the factory...
6 G-B	turn LH out (EWD 1993 p.130-131) / LH rear lamp R11 pin 6, socket pigtail - t...	**our w_rl_turn_l: 18 AWG Green** to OEM J/B 1 plug 1F pin 3 (LH kick), G-B wire (Turn LH rear +) EWD 1993 p.130-131 / 1993 p.130; crimp butt splice into the factory 1F-3 lead at J/B1...

RH rear lamp R12

R12 | device plug / lead tapped by our harness

1 G-W	stop RH	**our w_rl_stop_r12_c1: 16 AWG Green** to splice sp_rl_stop (Stop splice) (Stop RH +) EWD 1993 p.126; new-wire flying lead, soldered by Dan to the factory...
2 G	tail RH	**our w_rl_tail_r12_c2: 16 AWG Green** to splice sp_rl_tail (Tail splice) (Tail RH +) EWD 1993 p.113; new-wire flying lead, soldered by Dan to the factory...
3 G	rear side marker RH	**our w_rl_tail_r12_c3: 16 AWG Green** to splice sp_rl_tail (Tail splice) (Rear marker RH +) EWD 1993 p.113; new-wire flying lead, soldered by Dan to the factory...
4 R-B	back-up RH (Mitchell 1993 Fig. 16)	**our w_rl_rev_r12_c4: 16 AWG Red** to splice sp_rl_rev (Back-up splice) (Back-up RH +) EWD Mitchell 1993 Fig. 16 grid 63; 1992 p.132; new-wire flying lead, soldered by Dan to the factory...
5 W-B	lamp ground RH	**our w_rl_gnd_r12_c5: 16 AWG White** to splice sp_rl_gnd (Rear ground splice) (Rear lamp RH gnd) EWD 1993 p.113 / p.126; new-wire flying lead, soldered by Dan to the factory...
6 G-Y	turn RH out (EWD 1993 p.130-131) / RH rear lamp R12 pin 6, socket pigtail - t...	**our w_rl_turn_r: 18 AWG Green** to OEM J/B 1 plug 1F pin 2 (LH kick), G-Y wire (Turn RH rear +) EWD 1993 p.130-131 / 1993 p.130; crimp butt splice into the factory 1F-2 lead at J/B1...

Front wiper motor F9

F9 | device plug / lead tapped by our harness

1 L-O	wiper HIGH (EWD p.148) / Front wiper motor F9 pin 1 (cowl, driver side) - HIGH	**our w_wip_hi: 14 AWG Blue** to OEM IE1 pin 20 (left kick, cowl side) (Wiper HIGH) EWD 1993 p.148 / 1993 p.148; crimp butt splice into the factory cowl-side lead (ca...
2 L-B	wiper LOW (EWD p.148) / Front wiper motor F9 pin 2 (cowl, driver side) - LOW	**our w_wip_lo: 14 AWG Blue** to OEM IE1 pin 13 (left kick, cowl side) (Wiper LOW) EWD 1993 p.148 / 1993 p.148; crimp butt splice into the factory cowl-side lead (ca...
3 L-W	park switch S (EWD p.148) / OEM IE1 pin 6 (left kick, cowl side) - L-W, wiper...	**our w_wip_park: 14 AWG Blue** to OEM IE1 pin 6 (left kick, cowl side) (Wiper park sw) EWD 1993 p.148 / 1993 p.148; new-wire run terminated at the device's own mating pl...
4 L	WIPER 20 A out (EWD p.148) / Front wiper motor F9 pin 4 (cowl, driver side) - +B	**our w_wip_b: 14 AWG Blue** to OEM J/B No.1 1H pin 5 (left kick) (Wiper +B) EWD 1993 p.148 / 1993 p.148; crimp butt splice into the factory 1H-5 lead at J/B1...

LH retract motor R2

R2 | device plug / lead tapped by our harness

1 G-W	open (UP) limit	**our w_rtr_open_r2: 18 AWG Green** to splice sp_rtr_open (Retract open-limit join) (Retract open limit) EWD 1993 p.117; new-wire run terminated at the device's own mating pl...
2 G-O	LH retract motor pin 2 (EWD p.117) / LH retract motor R2 pin 2 - motor feed f...	**our w_rtr_r2_2: 16 AWG Green** to OEM IE1 pin 15 (left kick, cowl side) (LH retract motor 2) EWD 1993 p.117 / 1993 p.117; crimp butt splice into the factory cowl-side lead (ca...
3 G-Y	close (DOWN) limit	**our w_rtr_close_r2: 18 AWG Green** to splice sp_rtr_close (Retract close-limit join) (Retract close limit) EWD 1993 p.117; new-wire run terminated at the device's own mating pl...
4 G	motor feed from IE1 pin 14	**our w_rtr_r2_4_p: 16 AWG Green** to splice sp_rtr_r2_4 (LH retract motor feed) (LH retract motor 4) EWD 1993 p.117; new-wire run terminated at the device's own mating pl...
5 W-B	ground	**our w_rtr_gnd_r2: 16 AWG White** to splice sp_rtr_gnd_r2 (LH retract motor ground) (LH retract ground) EWD 1993 p.117; new-wire run terminated at the device's own mating pl...

RH retract motor R3

R3 | device plug / lead tapped by our harness

1 G-W	open (UP) limit	**our w_rtr_open_r3: 18 AWG Green** to splice sp_rtr_open (Retract open-limit join) (Retract open limit) EWD 1993 p.117; new-wire run terminated at the device's own mating pl...
2 G-B	RH retract motor pin 2 (EWD p.117) / RH retract motor R3 pin 2 - motor feed f...	**our w_rtr_r3_2: 16 AWG Green** to OEM IE1 pin 8 (left kick, cowl side) (RH retract motor 2) EWD 1993 p.117 / 1993 p.117; crimp butt splice into the factory cowl-side lead (ca...
3 G-Y	close (DOWN) limit	**our w_rtr_close_r3: 18 AWG Green** to splice sp_rtr_close (Retract close-limit join) (Retract close limit) EWD 1993 p.117; new-wire run terminated at the device's own mating pl...
4 G-Y	motor feed from IE1 pin 7	**our w_rtr_r3_4_p: 16 AWG Green** to splice sp_rtr_r3_4 (RH retract motor feed) (RH retract motor 4) EWD 1993 p.117; new-wire run terminated at the device's own mating pl...
5 W-B	ground	**our w_rtr_gnd_r3: 16 AWG White** to splice sp_rtr_gnd_r3 (RH retract motor ground) (RH retract ground) EWD 1993 p.117; new-wire run terminated at the device's own mating pl...

Spare | CAN | Shield drain | +12 V power | Sensor signal | Output / actuator | Ground / sensor return | Sensor supply +5/+8 V | Coil trigger | Injector drive | Other / lighting / misc

Fill = function (keyword rule on the wire-list description). Circle size = contact size (DRB102 only). Only the DRB102 is its real maker insert face; every other connector is a numbered grid with the true cavity count (pin order, not the maker face view). Wire lists are the truth. Spare = no wire on that cavity.

OEM JUNCTION BOXES AND SPLICE POINTS (3 of 4)

Washer motor W1

W1 | device plug / lead tapped by our harness

1 L-R	switch side (EWD p.148) / OEM washer W1 pin 1 cowl lead (firewall, cabin side...	**our w_wash_neg: 14 AWG Blue** to OEM washer W1 pin 1 cowl lead (firewall, cabin side) (Washer switch side) EWD 1993 p.148 / 1993 p.148; new-wire run terminated at the device's own mating pl...
1 L-R	switch side (EWD p.148) / OEM washer W1 pin 1 cowl lead (firewall, cabin side...	**our w_wash_neg: 14 AWG Blue** to Washer motor W1 pin 1 (RH front corner, low) (Washer switch side) EWD 1993 p.148 / 1993 p.148; new-wire run terminated at the device's own mating pl...
2 L	+12 V (EWD p.148) / Washer motor W1 pin 2 (RH front corner, low) - +12 V	**our w_wash_pos: 14 AWG Blue** to Washer motor W1 pin 2 (RH front corner, low) (Washer motor +12 V) EWD 1993 p.148 / 1993 p.148; crimp butt splice into the factory cowl-wire lead to...
2 L	+12 V (EWD p.148) / Washer motor W1 pin 2 (RH front corner, low) - +12 V	**our w_wash_pos: 14 AWG Blue** to OEM washer W1 pin 2 cowl lead (firewall, cabin side) (Washer motor +12 V) EWD 1993 p.148 / 1993 p.148; crimp butt splice into the factory cowl-wire lead to...

Old R/B No.2 position

left kick panel; relay block REMOVED, its factory leads stay in the loom

Leads / pins

2 W-L	door lock hot	**our w_rb2_wl: 14 AWG Red** to FB-LOGIC pin BUS1-6 (Door lock feed) EWD 1993 p.137 / p.170; crimp butt splice into the factory cowl-side W-L lead...
- L-Y	GAUGE 15A, ignition on	**our w_pwr_86_cp: 18 AWG Red** to POWER R6 pin 86 (Power relay coil+) EWD 1993 p.137-138; crimp butt splice into the factory cowl-side L-Y lead...
- W-G	power windows	**our w_rb2: 10 AWG Red** to POWER R6 pin 30 (Power window feed) EWD 1993 p.137; heavy crimp butt splice into the factory cowl-side W-...

Old R/B No.4 position

right kick panel; relay block REMOVED, its factory leads stay in the loom

Leads / pins

1 B-W	start	**our w_start_in_cp: 20 AWG Green** to FB-LOGIC pin BUS2-13 IN (Start sw ST1 feed) EWD 1993 TIS EWD p.66 (Starting and Ignition, All-Trac); solder splice onto the factory lead
- L-W	A/C amp HR	**our w_htr_85_cp: 18 AWG Violet** to POWER R1 pin 85 (Heater relay coil-) EWD 1993 p.215; crimp butt splice into the factory L-W lead at the ol...
- R-L	coil + feed	**our w_htr_86_cp: 18 AWG Red** to POWER R1 pin 86 (Heater relay coil+) EWD 1993 p.215; crimp butt splice into the factory R-L lead at the ol...

A/C amplifier

A18 | device plug / lead tapped by our harness

11 G-Y	A/C cancel (ACT)	**our w_ac_kill: 20 AWG Gray** to ECU-A pin A18 (A/C cancel (ACT)) EWD 1992 Celica EWD132U p.93 and p.218 (docs/electrical/ewd-snips/1992-celica-ewd132u-a17-act-p093.png, 1992-celica-ewd132u-a17-clutch-p218.png); 1990 ST185 book locator A18;...
13 Y-R	A/C pressure sw out	**our w_ac_psw_sw: 20 AWG Yellow** to splice sp_ac_psw (SP-AC-PSW: A/C pressure switch... (A/C pressure sw out) EWD 1990 ST185 EWD p.152 (docs/electrical/ewd-snips/1990-st185-autoac-pressure-clutch-p152.png); 1992 EWD132U p.219 A17-13...

A5

A5 | device plug / lead tapped by our harness

1 V-R	+12 V feed	**our w_acp_1: 18 AWG Violet** to FB-POWER pin ACPS (A/C press sw feed) EWD p.152; new-wire run terminated at the switch's own mating pl...
4 Y-R	A/C pressure sw out	**our w_ac_psw_in: 20 AWG Yellow** to splice sp_ac_psw (SP-AC-PSW: A/C pressure switch... (A/C pressure sw out) EWD 1990 ST185 EWD p.152 (docs/electrical/ewd-snips/1990-st185-autoac-pressure-clutch-p152.png); 1990 ST185 EWD p.152 (A5 pin 4 Y-R); new-wire run terminated at the switch's o...

cluster plug C11

C11 | device plug / lead tapped by our harness

2 G-B	turn LH indicator	**our w_tl_a: 20 AWG Green** to Turn LH LED (Turn LH LED +) EWD 1993 TIS EWD p.206 (Combination Meter; the 1993 book codes this plug C13 (B)); solder splice onto the factory lead at the combinatio...
11 G-Y	turn RH indicator	**our w_tr_a: 20 AWG Green** to Turn RH LED (Turn RH LED +) EWD 1993 TIS EWD p.206 (Combination Meter; the 1993 book codes this plug C13 (B)); solder splice onto the factory lead at the combinatio...
12 R-L	high beam indicator feed	**our w_hb_a: 20 AWG Red** to High beam LED (High beam LED +) EWD 1993 TIS EWD p.206 (Combination Meter; the 1993 book codes this plug C13 (B)); solder splice onto the factory lead at the combinatio...

cluster plug C12

C12 | device plug / lead tapped by our harness

1 R-G	park brake switch ground	**our w_pk_k: 20 AWG Red** to Park brake LED (Park brake sw (-)) EWD 1993 TIS EWD p.206 (Combination Meter; the 1993 book codes this plug C14 (C)); solder splice onto the factory lead at the combinatio...
8 Y	alternator L, charge lamp drive	**our w_chg_k: 20 AWG Yellow** to Charge LED (Charge lamp drive) EWD p.52; solder splice onto the factory lead at the combinatio...
9 B-O	IG +12 V feed for the LEDs	**our w_ig_tap: 20 AWG Black** to splice sp_led_12v (LED +12 V splice) (LED IG +12 V) EWD p.52; solder splice onto the factory lead at the combinatio...

cruise stalk switch C14

Cruise control switch (C14, comb. switch) | device plug / lead tapped by our harness

5 W	cruise ladder	**our w_an2: 20 AWG White** to splice sp_cruise_ladder (Cruise ladder splice) (Cruise ladder) EWD 1990 ST185 EWD p.116 (book p.118, connector C14 'cruise control SW [COMB. SW]'); 1992 Celica EWD132U p.180 same pins; 1993 Mitchell Electrical p.98-99 switch tests (docs/OPEN-ITEMS.md row 6); solder splice...
15 Y-B	cruise MAIN on	**our w_sw2: 20 AWG Green** to CSB3 pin 17 (Cruise MAIN on) EWD 1990 ST185 EWD p.116 (book p.118, connector C14 'cruise control SW [COMB. SW]'); 1992 Celica EWD132U p.180 same pins; 1993 Mitchell Electrical p.98-99 switch tests (docs/OPEN-ITEMS.md row 6); solder splice...
17 BR	stalk ground	**our w_cru_gnd: 20 AWG Black** to splice sp_csb_gnd (CSB3 ground splice) (Cruise stalk ground) EWD 1990 ST185 EWD p.116 (book p.118, connector C14 'cruise control SW [COMB. SW]'); 1992 Celica EWD132U p.180 same pins; 1993 Mitchell Electrical p.98-99 switch tests (docs/OPEN-ITEMS.md row 6); solder splice...

LH front turn / clearance lamp F7

F7 | device plug / lead tapped by our harness

1 G	clearance +	**our w_clr_f7: 16 AWG Green** to splice sp_clr (Clearance feed split) (LH clearance +) EWD 1993 p.112; new-wire run terminated at the device's own mating pl...
2 W-B	ground	**our w_f7_gnd: 16 AWG White** to Ground stud EB, LH front fender (EWD ground point... (LH front lamp gnd) EWD 1993 p.112 / p.130; new-wire run terminated at the device's own mating pl...
3 G-B	turn LH out (EWD p.130) / LH front turn / clearance lamp F7 pin 3 - turn +	**our w_turn_f7: 18 AWG Green** to OEM J/B No.1 1H pin 4 (left kick) (LH front turn +) EWD 1993 p.130 / 1993 p.130; crimp butt splice into the factory 1H-4 lead at J/B1...

Spare | CAN | Shield drain | +12 V power | Sensor signal | Output / actuator | Ground / sensor return | Sensor supply +5/+8 V | Coil trigger | Injector drive | Other / lighting / misc

Fill = function (keyword rule on the wire-list description). Circle size = contact size (DRB102 only). Only the DRB102 is its real maker insert face; every other connector is a numbered grid with the true cavity count (pin order, not the maker face view). Wire lists are the truth. Spare = no wire on that cavity.

OEM JUNCTION BOXES AND SPLICE POINTS (4 of 4)

RH front turn / clearance lamp F8

F8 | device plug / lead tapped by our harness

1	G	clearance +	**our w_clr_f8: 16 AWG Green** to splice sp_clr (Clearance feed split) (RH clearance +) EWD 1993 p.112; new-wire run terminated at the device's own mating pl...
2	W-B	ground	**our w_f8_gnd: 16 AWG White** to Ground stud EA, RH front fender (EWD ground point... (RH front lamp gnd) EWD 1993 p.112 / p.130; new-wire run terminated at the device's own mating pl...
3	G-Y	turn RH out (EWD p.130) / RH front turn / clearance lamp F8 pin 3 - turn +	**our w_turn_f8: 18 AWG Green** to OEM J/B No.1 1H pin 8 (left kick) (RH front turn +) EWD 1993 p.130 / 1993 p.130; crimp butt splice into the factory 1H-8 lead at J/B1...

LH headlamp H1

H1 | device plug / lead tapped by our harness

1	R-L	high-beam return	**our w_hl_high_h1: 16 AWG Red** to splice sp_hl_high (High-beam return join) (LH high-beam return) EWD 1993 p.116; new-wire run terminated at the device's own mating pl...
2	R-W	low-beam return	**our w_hl_low_h1: 16 AWG Red** to splice sp_hl_low (Low-beam return join) (LH low-beam return) EWD 1993 p.116; new-wire run terminated at the device's own mating pl...
3	R	+12 V feed	**our w_pmu_hl: 14 AWG Red** to LOGIC M1 pin 87 (Head LH +12 V out) EWD 1993 p.116; new-wire run terminated at the device's own mating pl...

RH headlamp H2

H2 | device plug / lead tapped by our harness

1	R-L	high-beam return	**our w_hl_high_h2: 16 AWG Red** to splice sp_hl_high (High-beam return join) (RH high-beam return) EWD 1993 p.116; new-wire run terminated at the device's own mating pl...
2	R-W	low-beam return	**our w_hl_low_h2: 16 AWG Red** to splice sp_hl_low (Low-beam return join) (RH low-beam return) EWD 1993 p.116; new-wire run terminated at the device's own mating pl...
3	R-Y	+12 V feed	**our w_pmu_hr: 14 AWG Red** to LOGIC M2 pin 87 (Head RH +12 V out) EWD 1993 p.116; new-wire run terminated at the device's own mating pl...

J/B No.3 (plug 3B)

behind the combination meter; OEM junction block KEPT

Plug 3B

9	R-L	GAUGE 15A, ignition on (Mitchell 1993 Fig. 16)	**our w_rev_gauge_cp: 16 AWG Red** to splice sp_rev_gauge (GAUGE 15A splice) (GAUGE 15A feed) EWD Mitchell 1993 overall wiring Fig. 16 (grid 63, J/B #3 pin B9); 1992 p.132; crimp butt splice into the factory R-L lead at J/B3 3...

IG2 plug

under the ECU, cowl side; engine-side wire removed, we tap the cowl-side leads

Leads / pins

15	B-R	ECU-IG 15 A feed	**our w_ig_ks_in: 16 AWG Red** to KILL SWITCH pin ALT 1 (Alt IG to kill sw) EWD 1993 p.74-75; crimp butt splice into the factory cowl-side lead at...
16	Y	charge lamp	**our w_alt_l_c: 20 AWG Yellow** to ALT PLUG G2 pin 2 (Alt L charge lamp) EWD 1993 p.74-75; crimp butt splice into the factory cowl-side lead at...

A/C compressor A4

A4 | device plug / lead tapped by our harness

4	B-W	magnet clutch +	**our w_acc_clutch: 16 AWG Red** to LOGIC M5 pin 87 (A/C clutch +12 V) EWD 1990 ST185 EWD p.152 (docs/electrical/ewd-snips/1990-st185-autoac-pressure-clutch-p152.png); 1992 EWD132U locator A4; new-wire run terminated at the device's own mating pl...

blower motor B4

B4 | device plug / lead tapped by our harness

2	B	blower +	**our w_rb4_cp: 8 AWG Red** to splice sp_htr87 (Heater relay output splice (4 AWG)) (Blower motor +) EWD 1993 p.215-216; heavy crimp butt splice into the factory lead at the...

clutch start switch C11

C11 clutch start switch (1993 code) | device plug / lead tapped by our harness

1	B	clutch pedal switch	**our w_sw5: 20 AWG Green** to CSB3 pin 20 (Clutch pedal sw) EWD 1993 TIS EWD p.66 (Starting and Ignition, All-Trac); solder splice onto the factory lead

High-mount stop light H6 plug A

H6 | device plug / lead tapped by our harness

1	G-W	HMSL +	**our w_rl_stop_h6_c1: 16 AWG Green** to splice sp_rl_stop (Stop splice) (High stop lamp +) EWD 1993 p.126; new-wire flying lead, soldered by Dan to the factory...
2	W-B	HMSL ground	**our w_rl_gnd_h6_c2: 16 AWG White** to splice sp_rl_gnd (Rear ground splice) (High stop lamp gnd) EWD 1993 p.126; new-wire flying lead, soldered by Dan to the factory...

Licence plate light L1

L1 | device plug / lead tapped by our harness

1	G	licence light +	**our w_rl_tail_l1_c1: 16 AWG Green** to splice sp_rl_tail (Tail splice) (Licence light +) EWD 1993 p.113; new-wire flying lead, soldered by Dan to the factory...
2	W-B	licence light ground	**our w_rl_gnd_l1_c2: 16 AWG White** to splice sp_rl_gnd (Rear ground splice) (Licence light gnd) EWD 1993 p.113; new-wire flying lead, soldered by Dan to the factory...

stop lamp switch S9

S9 | device plug / lead tapped by our harness

1	G-W	brake on, +12 V	**our w_sw6: 16 AWG Green** to OPTO8 pin IN1 (Brake on +12 V) EWD 1993 p.126-127; soldered tap into the factory G-W lead at S9 (cabin)
1	G-W	stop lamp feed	**our w_rl_stop: 16 AWG Green** to splice sp_rl_stop (Stop splice) (Stop lamp feed) EWD 1993 p.126-127; soldered tap into the factory G-W lead at S9 pin 1 (c...

Spare | CAN | Shield drain | +12 V power | Sensor signal | Output / actuator | Ground / sensor return | Sensor supply +5/+8 V | Coil trigger | Injector drive | Other / lighting / misc

Fill = function (keyword rule on the wire-list description). Circle size = contact size (DRB102 only). Only the DRB102 is its real maker insert face; every other connector is a numbered grid with the true cavity count (pin order, not the maker face view). Wire lists are the truth. Spare = no wire on that cavity.

REAR

IX-FUEL inline to fuel sender (trunk LH)

DT06-2S | RearFuel | 2 cavities | numbered grid (true count)

1	Fuel level signal
2	Fuel level ground

mates ix_fuel_r on B-cabin

FUEL LVL

DT04-2P | RearFuel | 2 cavities | numbered grid (true count)

1	Fuel level signal
2	Fuel level ground

FUEL PUMP

(pump pigtail - Walbro F90000295) | RearFuel | 2 cavities | numbered grid (true count)

1	Fuel pump +12 V
2	Fuel pump ground

WSS RL

DT06-2S-CE05 | WheelSpeed-Rear | 2 cavities | numbered grid (true count)

1	RL wheel speed +
2	RL wheel speed -

WSS RR

DT06-2S-CE05 | WheelSpeed-Rear | 2 cavities | numbered grid (true count)

1	RR wheel speed +
2	RR wheel speed -

IX-WS-R

DT04-6P | WheelSpeed-Rear | 6 cavities | numbered grid (true count)

1	VRC-R +5V
2	VRC-R GND
3	Rear left wheel spd
4	Rear right wheel spd
5	WS-R screen drain
6	

mates ix_ws_rear_p on B-cabin

VRC-R IN

LP-16-J05PE-02-001 | WheelSpeed-Rear | 5 cavities | numbered grid (true count)

1	RL wheel speed +
2	RL wheel speed -
3	Rear sensor shields
4	RR wheel speed +
5	RR wheel speed -

VRC-R OUT

LP-16-J05PE-02-001 | WheelSpeed-Rear | 5 cavities | numbered grid (true count)

1	VRC-R +5V
2	VRC-R GND
3	Rear left wheel spd
4	Rear right wheel spd
5	WS-R screen drain

Spare · CAN · Shield drain · +12 V power · Sensor signal · Output / actuator · Ground / sensor return · Sensor supply +5/+8 V · Coil trigger · Injector drive · Other / lighting / misc

Fill = function (keyword rule on the wire-list description). Circle size = contact size (DRB102 only). Only the DRB102 is its real maker insert face; every other connector is a numbered grid with the true cavity count (pin order, not the maker face view). Wire lists are the truth. Spare = no wire on that cavity.